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import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Step 1: Upload the image
uploaded = files.upload()

# Get the uploaded image name
image_path = next(iter(uploaded))

# Step 2: Read the image
image = cv2.imread(image_path)

# Check whether image is loaded
if image is None:
    print("Error: Image could not be loaded.")
else:
    # Step 3: Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Step 4: Calculate Sobel gradients
    gradient_x = cv2.Sobel(
        gray, cv2.CV_64F, 1, 0, ksize=3
    )

    gradient_y = cv2.Sobel(
        gray, cv2.CV_64F, 0, 1, ksize=3
    )

    # Step 5: Calculate gradient
    gradient = gradient_x - gradient_y

    # Convert result to displayable 8-bit image
    gradient = cv2.convertScaleAbs(gradient)

    # Step 6: Save the output
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output_path = "sharpened_image3.jpg"
cv2.imwrite(output_path, gradient)

# Step 7: Display images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(gradient, cmap="gray")
plt.title("Sobel Gradient Image")
plt.axis("off")

plt.show()

# Step 8: Download output
files.download(output_path)
```

Choose Files anime-boy-...ace-sign.jpg

anime-boy-giving-peace-sign.jpg(image/jpeg) - 9355745 bytes, last modified: 8/19/2026 - 100% done

Saving anime-boy-giving-peace-sign.jpg to anime-boy-giving-peace-sign.jpg

Original Image



Sobel Gradient Image

